

W3C LBD Community Group

Minutes - Call 16/06/2025

Attendees

- Katja Breitenfelder (Fraunhofer IBP, Germany)
- Alex Donkers (Eindhoven University of Technology, Netherlands)
- Mathias Bonduel (Neanex Technologies, Belgium)
- Marcel Durben (RWTH Aachen University, Germany)
- Irfan Čustović (Delft University of Technology, Netherlands)
- Wouter Lubbers (Semtech, Netherlands)
- Junsong Du (RWTH Aachen University, Germany)
- Avinash Ramachandruni (Eindhoven University of Technology, Netherlands)
- Ya Wen (University of Cambridge, UK)
- Karim Farghaly (UCL, UK)
- Conor Shaw (UCD, Ireland)
- Maxime Lefrançois (IMT - Mines Saint-Étienne, France)
- Frédéric Bosché (University of Edinburgh, UK)
- Arghavan Akbarieh (Eindhoven University of Technology, Netherlands)
- Eduard Loscos (Building Digital Twin Association, IDP, Spain)
- Joel J. Bender (Cornell, USA)
- Claudio Mirarchi (Polimi, Italy)
- Klaus Linhard (Hochschule München, IABI, Germany)
- Zhangcheng Qiang (Australian National university, Australia)
- Nicola Moretti (UCL, UK)
- Piotr Marek Smolira (DTU, Denmark)
- Pablo Vicente Legazpi (BDTA, Spain)
- Hervé Pruvost, Fraunhofer IIS, Germany)
- Michael Koster (KTC, Sweden).

Please join the W3C LBD CG and subscribe to the internal mailing list:

[Linked Building Data Community Group \(w3.org\)](https://www.w3.org/2025/06/16-lbd-minutes)

Links

- Presentation slides EC3: [link to Github](#)
- Presentation slides BDTA: [link to Github](#)
- Other interesting links:
 - EC3: Built Environment Ontology Lookup Service (BE-OLS)
 - web page: <https://cyberbuildlab.github.io/BE-OLS/>
 - Github: <https://github.com/CyberbuildLab/BE-OLS>
 - BDTA: <https://ontologies.buildingdigitaltwin.org/ontologies/#>

Date and time

- Monday 16th of June 2025, 15:00-16:30@UTC/ 16:00-17:30@CEST/ 07:00-08:30@PST

Moderators

1. Mathias Bonduel

Agenda

1. Introduction of new members
2. Presentation EC3
3. Presentation BDTA
4. Discussion
5. Further topics

Minutes

- **Introduction of new members**
 - Avinash Ramachandrani: PhD candidate, semantic consistency of data, working at TUE
 - Marcel Durben, RWTH Aachen, 1st year PhD student, background mechanical engineering, focus HVAC
 - Eduard Loscos, president BDTA, will be presenting
 - Ya Wen, PhD holder from Univ. of Cambridge, topic linked to facility and asset Management, now searching for more practical application of previous work
 - Frederic Bosche, team leader, data ontologies, interested in harmonizing all initiatives in the area
- **EC3 presentation (Arghavan Akbarieh)**
 - European Council on Computing in Construction: “**EC3 Built Environment - Ontology Lookup Service (BE-OLS)**”
 - Arghavan (presenter), postdoc researcher, TUE, part of EC3
 - Modelling & Standards (M&C) Committee, open to new members
 - [EC3 - BIM Standards Landscape Explorer](#)
 - EC3 - BE-OLS
 - facilitating (re)use of existing ontologies
 - allows the search for available ontologies
 - categorized and analyzed catalogued ontologies > 100 ontologies
 - bundled under domains, 20 subdomains
 - analytics with regard to categorization (domains etc.), networks (how ontologies are related to other domain or non-domain ontologies)
 - metadata and enriched metadata of ontologies available
 - -> How to create a rating system for ontologies, and which frameworks exist already?
 - **A complementary evaluation framework for ontology (re)use**
 - Works started in October 2024, carried out by 5 academic volunteers
 - Considering FAIS Score
 - EC3 evaluation on ‘**Connectivity**’ (to upper ont., to existing AECO ont., to meta schema ont.), ‘**Accessibility**’ (as a data model, as a serialisation, as a URI), and ‘**Documentation & Reuse**’ (clear natural language description in URI or serialization, proper use of annotations, reused/extended).
 - Ongoing work, scope of evaluation to be widened
 - Future Work: completing missing ontologies, enhancing information quality, completing evaluations on ‘Quantity’ and ‘Robustness’, Extracting metadata from ttl files (to check the overlap of concepts), using LLMs

- New Automated Ontology Submission System, open for submission of ontologies by the audience.
- Recommendations (Maxime): Recommendations for assessing quality and fairness of ontos:
 - Evaluation of the OQuaRE framework for ontology quality, ---> <https://www.cs.man.ac.uk/~stevensr/papers/OQuareProof.pdf>
 - O'FAIRe makes you an offer, DOI 10.1504/IJMSO.2022.10054778
 - 'Modularity' (ex. there could be only one entry for SOSA/SSN, the COGITO ontologies / BIMERR / SAREF and extensions)
 - 'Versioning' (there are different versions of SAREF, each may have different scores for accessibility etc. and in the future you may want to add a new version of an existing ontology)
 - Results of the FAIR IMPACT project: <https://cordis.europa.eu/project/id/101057344/results>
 - Deploy an instance of OntoPortal instead, and extend with whatever is not already supported by OntoPortal. Join the alliance
- Recommendations (Mathias): Check 'Linked Open Vocabularies'; <https://ontoportal.org/about/>
- Recommendation (Alex): <https://agroportal.lirmm.fr/>. One of the examples, an agricultural ontology repository using ontoportal
- Recommendations (Maxime): Example of an instance of OntoPortal: <https://technoportal.hevs.ch/>, <https://ontoportal.org/about/>; BTW about storing versions of existing ontologies: LOV actually has ontology/ontology versions that would otherwise have disappeared from the web, like <https://lov.linkeddata.es/dataset/lov/vocabs/acco> -> 404 on <http://purl.org/acco/ns>, but LOV has snapshots of v0.1 and v1.0, example <https://lov.linkeddata.es/dataset/lov/vocabs/acco/versions/2013-03-25.n3>
- Q (Wouter): Question on accessibility: problem that ontologies are only released at files, does 'as URI' mean it is accessible via SPARQL endpoint. A (Arghavan): currently not evaluated.
- A (Arghavan): Only listing of ontos, possible include versioning / updates of ontos, but no hosting. (Frederic): Maintenance of ontos is the task of the owners.
- **BDTA presentation (Eduard Loscos)**
 - BDTA strategy, includes 5 WGs:
 - **WG Definitions and Roles**
 - 2019 BDTA whitepaper no. 1: providing first definitions on digital twins in the built environment, outcome of EU project SPHERE
 - Related Committees and Publications:
 - CEN TEchnical Committee 442 (BIM)
 - WI 00442049 CEN/TR 18077:2024: Digital twins in built environment **Use Cases** (Case Studies), clustering important for later ontology catalogue
 - WI 00442055 prEN 18162 Building information modeling
 - WG2 Open Technology Framework
 - whitepaper no. 3 'ontology networks for DTs' (2022)
 - 1st **BDTA Roadmap** agreed in BDTIC4, Barcelona 2024
 - **Ontology Network Catalogue**
 - Phase 1 (2025) finished: *Directory of Ontologies in the BDT domain*
 - Facilitating easy search of ontos
 - 3 clear application cases, to be standardised

- **1 - Operation & Maintenance (O&M)** Optimisation and linking with time series data
- **2 - Prototype Optimisation**
- **3 - Digital Building Permits**
- Outlook: application case Digital Product Passports
- Start of Phase 2: *Toolkit and guidelines*
- Onto adoption and interaction
- Guidelines and tutorials
- BDTA's Future Activities
 - Alignment and definition of Interoperable Ontologies principles
 - Improved interoperability of existing Domain Ontos
 - Catalogue of BDT Ontos
- Sample catalogue entry:

Digital Construction Ontologies (DiCon)

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Ontology name	Digital Construction Ontologies (DiCon)
Release date	01/07/2022
Description	Digital Construction Ontology - semantic interoperability between systems in the construction and renovation domain. They define the basic terms related to built assets, building design, construction project planning, and construction planning, including the agents, resources and spatio-temporal aspects. The definitions rely heavily on standards (ISO 19650 BIM/IM, ISO 16739-1 IFC, ISO/IEC 21838-2 BFO and ISO 21597-1 ICDD) and on established ontologies (such as OWL-Time, PROV-O and SSN/SOSA). The correspondences of Digital Construction Ontologies with other ontologies are provided explicitly in separate alignment modules.
License	CC BY 4.0
DT Sub-domain	Building Digital Twin Operation & Maintenance (O&M) Optimisation
Website URI	https://digitalconstruction.github.io/
Publisher	Seppo Törmä
Author(s)	BIM4EEB, DICTION EU PROJECTS
Language	English
Version	0.5
Ontology Language	OWL
Use Cases	https://www.bim4eeb-project.eu/demo-cases.html

- Q (Mathias): BDTA has more industry members. Recommendation: Not only track related use cases, but would be interesting to cover which ontos are widely used (by industry) which are not, and for what ontos are used. Experience and success stories by the industry. That would be valuable. A (Eduard): Bottom-up approach. Recalling last BDTA WG meetings collecting 10 EU projects, all of them dealing with semantics (equivalent to approx. 60 Mill. €), shows the relevance. In this connection with industry it is indeed important to figure out which ontologies work / had worked best, to propose it as a future standard.
- Q (Joel): Similar initiatives in the US:
 - <https://www.linuxfoundation.org/press/intent-to-form-c4sb-foundation>
 - <https://www.c4sb.org>

- **Discussion**

- Q (Irfan): underlying datasets, it can become difficult to assess the performance, are there ideas to link ontos to datasets in order to provide evidence for the practice? Q (Mathias): Catalogues could facilitate access to datasets as well. A (Arghavan): We cover that aspect with the category 'robustness'. which ontologies to use for which use case is relevant. Open to hints on how to include this aspect. Community is invited to review. Generally an ontology can be assigned to multiple domains and subdomains.
- Q (Karim): From his experience bottom-up approaches do not work. More important to agree on why (then on what). No Competency Questions (CQs). Don't we have to shift now to a clear why (ontology development, classes etc.)? A (Eduard): We are linking ontos to use cases, with the clear vision of standardisation! A (Mathias): structuring and publishing ontos well, is important. Covering CQs would be important in the future.
- Q (Avinash): What kind of metrics do we have to look for when choosing an ontology for a specific application? A (Eduard): "Metrics not considered, but answering the question of "why" was considered.
- Remark (Avinash): When starting one year ago, we had no clear vision. In the upcoming year, it should be possible to develop KPIs and metrics. A (Eduard): That is very good, we can underpin our work with a more academic approach.
- Q (Piotr): Question on ontology alignment. A (Mathias): you can do alignment within the ontology, or you can publish alignments separately.

- **Further topics --**

Next Call

- **Summer Break: July and August 2025**
- **Next Call: 15/09/2025**, Monday, 15:00-16:30@UTC/ 16:00-17:30@CET/ 07:00-08:30@PST

Agenda: TBD

We are interested in getting suggestions from the community about potential agenda items and **Elevator Pitches** for the following calls. Please send your suggestions to the chairs or to internal-lbd@w3.org, whether you have a short presentation to bootstrap the discussion, and an approximate duration you think the discussion will last.

Previous minutes

<https://github.com/w3c-lbd-cg/lbd/tree/gh-pages/minutes>